SLIIT

COMPUTING | BUSINESS | ENGINEERING

IE3140

Advanced Network Engineering

3rd Year, Semester I

Submitted to

Sri Lanka Institute of Information Technology

In partial fulfillment of the requirements for the

Bachelor of Science Special Honors Degree in Information Technology

IT Number	Name
Amaraguru. M. S. B	**IT24102559**
Jayantha. K. P	**IT24103400**
Fernando. L. A. S	**IT24103080**
Sathsaranee. K. L. U	**IT24104150**

Section	Description
1.	Requirements and Assessment Alignment
2.	Network Architecture and Device Inventory
3.	IPv4 VLSM Calculations
4.	IPv6 Addressing and Dual Stack
5.	VLANs, Trunks and Inter-VLAN Routing
6.	DHCP Servers and Endpoint Addressing
7.	NAT/PAT Configuration and Verification
8.	Single-Area OSPF and Authentication
9.	Multi-Area OSPF Design
10.	Layer 2 Security Implementation
11.	Wireless and Service Reachability
12.	Multicast Design and Implementation Limitations
13.	Troubleshooting and Recovery
13.1.	Recovery and Root Cause Analysis
14.	Verification Results and Remaining Evidence
15.	Corrective Actions and Configuration Persistence
16.	Conclusions and Recommendations
17.	References

1.Requirements and assessment alignment

The TechNova Solutions enterprise network project was designed and implemented using Cisco Packet Tracer to meet the requirements of the IE3140 Advanced Network Engineering assignment. The project covers IPv4 VLSM addressing, IPv6 dual-stack implementation, VLAN configuration, inter-VLAN routing, Layer 2 security, dynamic routing using OSPFv2 and OSPFv3, and Network Address Translation using PAT.

The network consists of three office floors, a server farm, and a remote branch connected through a core network and an ISP. The implementation was verified using router and switch configuration commands, DHCP allocation results, routing tables, NAT translations, and end-to-end connectivity tests.

Table 2. Requirements traceability

Task	Required Outcome	Project Implementation and Evidence
A1	IPv4 VLSM addressing and subnet calculations	Completed VLSM design using 192.168.1.0/24 for Engineering, HR, Sales, Branch, Servers, Management, and WAN links.
A2	IPv6 addressing and dual-stack implementation	IPv6 /64 subnets configured for departmental networks, branch, and router links using 2001:DB8:ACAD::/48. IPv6 connectivity successfully verified.
A3	NAT overload (PAT) and translation verification	PAT configured on EDGE-ROUTER using public IPv4 address 203.0.113.5. Successful ISP connectivity and NAT translation entries verified.
B1	VLAN configuration and trunking	VLANs 10, 20, 30, 40, 99, and native VLAN 999 configured. Departmental VLAN membership and 802.1Q trunks

		verified.
B2	Inter-VLAN routing	Router-on-a-stick implemented using router subinterfaces. Communication between HR, Engineering, Sales, Branch, and Servers successfully tested.
B3	Port Security, BPDU Guard, and DHCP Snooping	Port Security with sticky MAC addresses and DHCP Snooping verified on departmental and branch switches. BPDU Guard configuration requires final evidence verification.
C1	Single-area OSPFv2 with MD5 authentication	OSPFv2 routing implemented and inter-router connectivity verified. MD5 authentication configuration requires final documentation.
C2	Multi-area OSPF routing	OSPF areas 0, 1, 2, and 3 implemented. Inter-area IPv4 routes verified, and OSPFv3 neighbor relationships and IPv6 routes successfully tested.
C3	Multicast routing	Excluded from implementation following the lecturer's instruction because the selected Packet Tracer router models do not support the required multicast functionality.
C4	OSPF troubleshooting and verification	Routing tables, OSPF neighbor states, and end-to-end connectivity verified. A documented fault diagnosis and recovery demonstration remains to be included.

The implementation demonstrates successful IPv4 and IPv6 communication between the enterprise departments, server farm, branch network, and ISP. Centralized DHCP services provide dynamic IPv4 addresses, while Layer 2 security mechanisms protect access-switch ports against unauthorized devices and DHCP-related attacks.

Multicast Scope Adjustment: The lecturer instructed students not to implement the multicast component because of limitations associated with the selected Cisco Packet Tracer router models. Therefore, multicast configuration and testing were excluded from the implemented project.

The report documents the network design, configurations, verification commands, testing results, and supporting screenshots to demonstrate the functionality of the completed network.

2. Network Architecture and Device Inventory

The TechNova Solutions enterprise network was designed using a hierarchical routed architecture to connect three office floors, a centralized server farm, and a remote branch. The topology consists of an EDGE router, a CORE router, departmental area routers, a branch router, an ISP router, and multiple access switches.

The EDGE router provides external connectivity and performs IPv4 Network Address Translation using PAT. The CORE router acts as the OSPF backbone, connecting the departmental networks and the EDGE router. The Area 1 router connects the Human Resources and Engineering departments, while the Area 2 router connects Sales and the centralized Server Farm. The branch router provides connectivity to the remote office through a serial WAN connection.

Departmental segmentation is implemented using VLANs, while inter-VLAN routing is performed through router subinterfaces. OSPFv2 provides dynamic IPv4 routing, and OSPFv3 supports IPv6 routing between the enterprise networks.

Table 3. Network Infrastructure Inventory

Device	Model / Type	Role and Connections
EDGE-ROUTER	Cisco Router	Connects CORE to ISP; performs NAT/PAT
CORE_Router	Cisco Router	OSPF backbone; connects EDGE, Area 1 and Area 2
Area01_R02	Cisco Router	Connects HR and Engineering VLANs to CORE
AREA-02-ROUTER	Cisco Router	Connects Sales, Server Farm and Branch WAN
BRANCH-ROUTER	Cisco Router	Connects remote Branch LAN through serial WAN
R1-HQ (ISP)	Cisco Router	Simulated ISP network and external connectivity
Floor_01_SW01	Cisco Switch	HR access switch; VLAN 10
Floor_02_SW02	Cisco Switch	Engineering access switch; VLAN 20
FLOOR_03_SW04	Cisco Switch	Sales access switch; VLAN 30
Branch_SW05	Cisco Switch	Branch LAN access switch

Server Farm Switch	Cisco Switch	Connects DHCP server and other server devices; VLAN 40
DHCP Server	Server-PT	Centralized IPv4 address allocation and DHCP services
Department PCs	PC-PT	End-user devices in HR, Engineering and Sales
Branch PCs	PC-PT	End-user devices in the remote branch

Table 4. Departmental Network and Equipment Inventory

Location / Network	Implemented Devices	Network Configuration
Floor 1 – Human Resources	1 access switch and 4 verified PCs	VLAN 10; 192.168.1.64/27
Floor 2 – Engineering	1 access switch and 3 verified PCs	VLAN 20; 192.168.1.0/26
Floor 3 – Sales	1 access switch and 3 verified PCs	VLAN 30; 192.168.1.96/27
Remote Branch	1 access switch and 3 verified PCs	Branch LAN; 192.168.1.128/27
Server Farm	Server switch and centralized DHCP server	VLAN 40; 192.168.1.160/28
Management Network	Network device management interfaces	VLAN 99; 192.168.1.176/29
External Network	EDGE router and ISP router	IPv4 203.0.113.4/30; IPv6 2001:DB8:ACAD:105::/64
Core Network	CORE, EDGE and area routers	OSPFv2 and OSPFv3 routing

2.1 Physical topology and forwarding paths

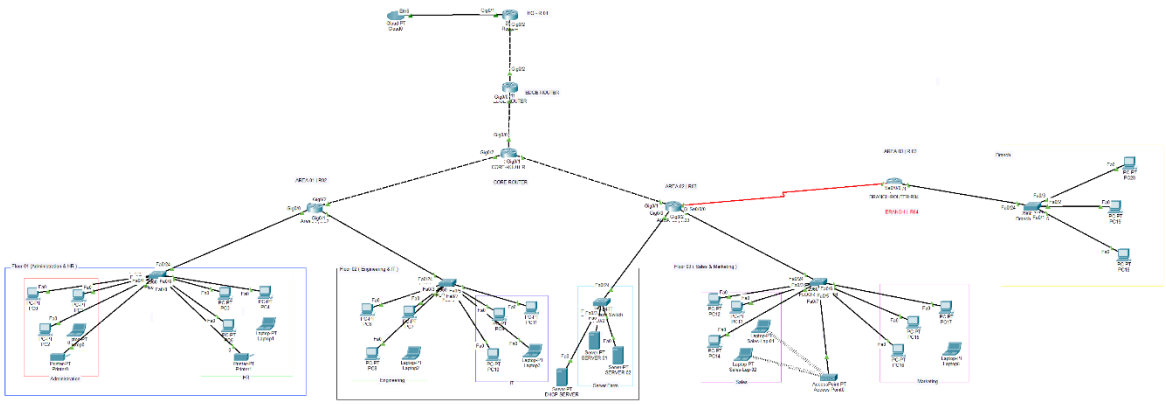


Table 5. Router and Switch Uplink Connections

Endpoint A	Endpoint B	Connection Purpose
EDGE-ROUTER G0/0	CORE_Router G0/0	Backbone link – 192.168.1.192/30
CORE_Router G0/2	Area01_R02 G0/2	Area 1 backbone – 192.168.1.184/30
CORE_Router G0/1	AREA-02-ROUTER G0/1	Area 2 backbone – 192.168.1.188/30
AREA-02-ROUTER S0/0/0	BRANCH-ROUTER S0/0/0	Branch serial WAN – 192.168.1.196/30
Area01_R02 G0/0	Floor_01_SW01 Fa0/24	HR VLAN 10 trunk; native VLAN 999
Area01_R02 G0/1	Floor_02_SW02 Fa0/24	Engineering VLAN 20 trunk; native VLAN 999
AREA-02-ROUTER G0/2	FLOOR_03_SW04 Fa0/24	Sales VLAN 30 trunk; native VLAN 999
AREA-02-ROUTER G0/0	Server Farm Switch	Server VLAN 40 trunk; native VLAN 999
BRANCH-ROUTER G0/1	Branch_SW05 Fa0/24	Branch LAN access connection
EDGE-ROUTER G0/2	R1-HQ (ISP) G0/2	External IPv4 203.0.113.4/30 and IPv6 2001:DB8:ACAD:105::/64

2.2 Network Forwarding Paths

Engineering to Server Farm

Engineering PC → Floor_02_SW02 → Area01_R02 → CORE_Router → AREA-02-ROUTER → Server Farm Switch → Server

Branch to ISP

Branch PC → Branch_SW05 → BRANCH-ROUTER → AREA-02-ROUTER → CORE_Router → EDGE-ROUTER → ISP

HR to Sales

HR PC → Floor_01_SW01 → Area01_R02 → CORE_Router → AREA-02-ROUTER → FLOOR_03_SW04 → Sales PC

3. IPv4 VLSM Calculations

The TechNova Solutions enterprise network uses the private IPv4 address block 192.168.1.0/24, which provides 256 total IPv4 addresses. Variable Length Subnet Masking (VLSM) was applied to allocate appropriately sized subnets to Engineering, Human Resources, Sales, the remote Branch, the Server Farm, Management, and router-to-router WAN links.

Table 6. Subnet Calculations and Host Capacity

Segment	Required Hosts	Host Bits	Calculation	Prefix	Block Size
Engineering	50	6	64 − 2 = 62	/26	64
HR	30	5	32 − 2 = 30	/27	32
Sales	25	5	32 − 2 = 30	/27	32
Branch	20	5	32 − 2 = 30	/27	32
Server Farm	10	4	16 − 2 = 14	/28	16
Management	5	3	8 − 2 = 6	/29	8
Each Router-to-Router Link	2	2	4 − 2 = 2	/30	4

3.1 Worked Example and Allocation Boundaries

The Engineering department requires 50 usable IPv4 addresses. Five host bits provide only 30 usable addresses, which is insufficient. Therefore, six host bits are required, providing 62 usable addresses.

The Engineering subnet uses a /26 prefix, corresponding to the subnet mask 255.255.255.192**.**

Table 7. IPv4 VLSM Address Allocation

Segment	Network Address	Subnet Mask	Usable Host Range	Broadcast	Gateway
Engineering	192.168.1.0/26	255.255.255.192	192.168.1.1–62	192.168.1.63	192.168.1.1
HR	192.168.1.64/27	255.255.255.224	192.168.1.65–94	192.168.1.95	192.168.1.65
Sales	192.168.1.96/27	255.255.255.224	192.168.1.97–126	192.168.1.127	192.168.1.97
Branch	192.168.1.128/27	255.255.255.224	192.168.1.129–158	192.168.1.159	192.168.1.129
Server Farm	192.168.1.160/28	255.255.255.240	192.168.1.161–174	192.168.1.175	192.168.1.161
Management	192.168.1.176/29	255.255.255.248	192.168.1.177–182	192.168.1.183	192.168.1.177
Area 1 WAN	192.168.1.184/30	255.255.255.252	192.168.1.185–186	192.168.1.187	N/A
Area 2 WAN	192.168.1.188/30	255.255.255.252	192.168.1.189–190	192.168.1.191	N/A
EDGE–CORE WAN	192.168.1.192/30	255.255.255.252	192.168.1.193–194	192.168.1.195	N/A
Branch WAN	192.168.1.196/30	255.255.255.252	192.168.1.197–198	192.168.1.199	N/A

3.2 IPv4 Address Utilization

The enterprise network allocates addresses to six LAN subnets and four point-to-point WAN subnets**.**

Table 8. IPv4 Address Utilization

Measure	Value	Description
Original address block	192.168.1.0/24	256 total IPv4 addresses
Allocated addresses	200	Total addresses allocated to 10 subnets
Usable host addresses	180	Excludes network and broadcast addresses
Network addresses	10	One per subnet
Broadcast addresses	10	One per subnet
Unallocated addresses	56	Available for future expansion
Unallocated address range	192.168.1.200–192.168.1.255	Remaining private address space
Address utilization	78.125%	Percentage of allocated address space
Remaining address capacity	21.875%	Percentage available for expansion

3.3 Addressing Efficiency and Future Expansion

The VLSM design improves IPv4 address utilization by allocating subnet sizes according to departmental host requirements. Engineering receives a /26 subnet, while HR, Sales, and Branch each receive /27 subnets. Smaller /28 and /29 subnets are used for the Server Farm and Management network, respectively.

Point-to-point router links use /30 subnets, providing exactly two usable host addresses per link. The remaining 56 IPv4 addresses, from 192.168.1.200 to 192.168.1.255, are reserved for future network expansion.

The implemented addressing scheme provides sufficient capacity for the stated host requirements while maintaining a structured and scalable network design.

4. IPv6 addressing and dual stack

The TechNova Solutions enterprise network implements IPv6 addressing alongside IPv4 to support dual-stack communication across departmental networks, the centralized Server Farm, the remote Branch, and the simulated ISP.

The IPv6 addressing scheme uses the documentation prefix 2001:DB8:ACAD::/48, which provides 65,536 possible /64 subnets. Each departmental LAN is assigned a separate /64 prefix, while dedicated /64 subnets are allocated to router-to-router links.

IPv6 routing is implemented using OSPFv3, allowing communication between different IPv6 networks. Successful OSPFv3 neighbor relationships, inter-area IPv6 routes, and end-to-end connectivity tests were verified using Cisco Packet Tracer.

Table 10. IPv6 Addressing Plan and Verified Configuration

Network Segment	IPv6 Subnet Prefix	Router Address / Example	Status
Human Resources (VLAN 10)	2001:DB8:ACAD:10::/64	2001:DB8:ACAD:10::1	Configured and tested
Engineering (VLAN 20)	2001:DB8:ACAD:20::/64	2001:DB8:ACAD:20::1	Configured and tested
Sales (VLAN 30)	2001:DB8:ACAD:30::/64	2001:DB8:ACAD:30::1	Configured and tested
Server Farm (VLAN 40)	2001:DB8:ACAD:40::/64	2001:DB8:ACAD:40::1	Configured and tested
Branch LAN	2001:DB8:ACAD:50::/64	2001:DB8:ACAD:50::1	Configured and tested
CORE–Area 1	2001:DB8:ACAD:101::/64	CORE ::1, Area 1 ::2	Configured
CORE–Area 2	2001:DB8:ACAD:102::/64	CORE ::1, Area 2 ::2	Configured
CORE–EDGE	2001:DB8:ACAD:103::/64	CORE ::1, EDGE ::2	Configured
Area 2–Branch WAN	2001:DB8:ACAD:104::/64	Area 2 ::1, Branch ::2	Configured
EDGE–ISP	2001:DB8:ACAD:105::/64	EDGE ::1, ISP ::2	Configured and tested

The IPv6 address allocation separates departmental and WAN traffic into independent /64 networks. This approach supports structured addressing, route identification, and future expansion.

The Management VLAN (VLAN 99) is configured for IPv4 management purposes. A separate IPv6 management prefix has not been confirmed in the verified configuration.

4.1 IPv6 verification evidence

```
CORE_Router>enable
CORE_Router#show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
5.5.5.5           1   FULL/DR         00:00:32    1               GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:32    2               GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:32    3               GigabitEthernet0/2
CORE_Router#show ospf ro
CORE_Router#show ospf rou
CORE_Router#show ospf route
                ^
% Invalid input detected at '^' marker.

CORE_Router#show ip ospf route
                   ^
% Invalid input detected at '^' marker.

CORE_Router#show ipv6 rou
CORE_Router#show ipv6 route
IPv6 Routing Table - 14 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OE2 ::/0 [110/1]
     via FE80::2D0:58FF:FE96:A901, GigabitEthernet0/0
OI  2001:DB8:ACAD:10::/64 [110/2]
     via FE80::260:5CFF:FE4E:2103, GigabitEthernet0/2
OI  2001:DB8:ACAD:20::/64 [110/2]
     via FE80::260:5CFF:FE4E:2103, GigabitEthernet0/2
OI  2001:DB8:ACAD:30::/64 [110/2]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
OI  2001:DB8:ACAD:40::/64 [110/2]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
OI  2001:DB8:ACAD:50::/64 [110/66]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
C   2001:DB8:ACAD:101::/64 [0/0]
     via GigabitEthernet0/2, directly connected
 --More--
```

Copy Pa

The screenshot verifies successful OSPFv3 routing on the CORE router. Three neighboring routers (5.5.5.5, 3.3.3.3, and 2.2.2.2) have established FULL/DR adjacencies. The IPv6 routing table shows learned inter-area routes (OI) to the HR, Engineering, Sales, Server Farm, and Branch networks, along with an external default route (OE2), confirming successful IPv6 route exchange across the enterprise network.

4.2 Dual-Stack Configuration

Dual-stack networking allows IPv4 and IPv6 to operate simultaneously on the same router interface. In this project, departmental router subinterfaces were configured with both IPv4 and IPv6 addresses.

For example, the HR VLAN 10 subinterface on the Area 1 router was configured as follows:

```
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.65 255.255.255.224
 ip helper-address 192.168.1.162
 ip ospf 1 area 1
 ipv6 address 2001:DB8:ACAD:10::1/64
 ipv6 ospf 10 area 1
!
```

The interface uses IPv4 address 192.168.1.65/27 and IPv6 address 2001:DB8:ACAD:10::1/64, allowing devices in the HR department to communicate using either protocol.

The encapsulation dot1Q 10 command associates the subinterface with VLAN 10, while ip helper-address forwards IPv4 DHCP requests to the centralized DHCP server.

5. VLANs trunks and inter VLAN routing

The TechNova Solutions enterprise network implements Virtual Local Area Networks (VLANs) to separate departmental traffic and improve network organization and security. The configured VLANs include HR (VLAN 10), Engineering (VLAN 20), Sales (VLAN 30), Servers (VLAN 40), Management (VLAN 99), and Native VLAN 999.

Departmental access switches connect end-user devices to their assigned VLANs. IEEE 802.1Q trunk links carry VLAN traffic between the access switches and departmental routers. Inter-VLAN routing is implemented using the router-on-a-stick technique, where router subinterfaces act as the default gateways for individual VLANs.

The remote Branch uses a separate LAN subnet connected through the Branch router rather than a departmental 802.1Q trunk.

Table 11. VLAN Purpose and Access Port Assignments

VLAN	Purpose	Configured Switch Ports / Usage
10	Human Resources	Floor_01_SW01 Fa0/5–Fa0/8; four HR PCs
20	Engineering	Floor_02_SW02 Fa0/1–Fa0/3; three Engineering PCs
30	Sales	FLOOR_03_SW04 Fa0/1–Fa0/3; three Sales PCs
40	Server Farm	Server Farm switch; centralized DHCP server and server devices
99	Management	Reserved for network device management
999	Native VLAN	Configured as the native VLAN on departmental 802.1Q trunks

Table 12. Verified VLAN Trunk Configuration

Switch / Trunk Port	Connected Router Interface	VLANs Allowed on Trunk	Native VLAN
Floor_01_SW01 Fa0/24	Area01_R02 G0/0	10, 99, 999	999
Floor_02_SW02 Fa0/24	Area01_R02 G0/1	20, 99, 999	999
FLOOR_03_SW04 Fa0/24	AREA-02-ROUTER G0/2	30, 99, 999	999
Server Farm switch uplink	AREA-02-ROUTER G0/0	VLAN 40; other allowed VLANs require confirmation	999 (planned configuration)

```
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.65 255.255.255.224
 ip helper-address 192.168.1.162
 ip ospf 1 area 1
 ipv6 address 2001:DB8:ACAD:10::1/64
 ipv6 ospf 10 area 1
!
```

The encapsulation dot1Q 10 command associates the subinterface with HR VLAN 10. The IPv4 address 192.168.1.65/27 serves as the default gateway for HR devices, while the IPv6 address 2001:DB8:ACAD:10::1/64 supports dual-stack communication.

The ip helper-address 192.168.1.162 command forwards DHCP requests from HR devices to the centralized DHCP server in the Server Farm.

6. DHCP servers and endpoint addressing

The TechNova Solutions enterprise network uses a centralized Dynamic Host Configuration Protocol (DHCP) server to automatically assign IPv4 addresses to end-user devices across the HR, Engineering, Sales, and remote Branch networks.

The DHCP server is located in the Server Farm (VLAN 40) and uses the static IPv4 address 192.168.1.162/28, with the default gateway 192.168.1.161. Departmental networks use separate VLSM subnets, and DHCP pools are configured according to their respective subnet masks and gateway addresses.

Since the DHCP server is located in a different subnet from most client devices, DHCP relay is implemented on the relevant router interfaces using the ip helper-address command. This enables DHCP broadcast requests to be forwarded to the centralized server across the routed enterprise network.

Table 13. DHCP Pool Configuration and Address Allocation

DHCP Pool	Starting IP Address	Subnet Mask	Default Gateway	Maximum Users
ENG_POOL	192.168.1.2	255.255.255.192	192.168.1.1	61
HR_POOL	192.168.1.66	255.255.255.224	192.168.1.65	29
SALES_POOL	192.168.1.98	255.255.255.224	192.168.1.97	29
BRANCH_POOL	192.168.1.130	255.255.255.224	192.168.1.129	29

Each DHCP pool is configured with a valid subnet mask, default gateway, and address range. Gateway addresses are excluded from the DHCP allocation ranges to avoid address conflicts.

The DHCP server also contained a default serverPool entry during configuration. This pool should be reviewed and removed or disabled if it is not required, particularly if its address range overlaps other networks.

Printers and Name Resolution

The TechNova Solutions network uses a centralized DHCP server at 192.168.1.162 to provide IPv4 addressing information to the HR, Engineering, Sales, and Branch networks. DHCP relay is configured on the departmental router interfaces to forward client requests to the centralized server.

The HR department uses VLAN 10 with subnet 192.168.1.64/27 and default gateway 192.168.1.65. If network printers are included, they should be assigned static IPv4 addresses outside the active DHCP allocation range to prevent address conflicts. Printer connectivity and addressing require separate verification.

The verified HR, Engineering, and Sales clients use 8.8.8.8 as their configured DNS server, while Branch clients receive 192.168.1.162. However, successful IP connectivity does not confirm DNS name resolution. A functioning DNS service and appropriate DNS records are required for hostname-based communication.

DHCP Relay Configuration

The following commands represent the DHCP relay configuration for the departmental networks:

```
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.1.161 255.255.255.240
 ip ospf 1 area 2
 ipv6 address 2001:DB8:ACAD:40::1/64
 ipv6 ospf 10 area 2
!
interface GigabitEthernet0/0.999
 encapsulation dot1Q 999 native
 no ip address
!
interface GigabitEthernet0/1
 ip address 192.168.1.190 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 cisco3140
 ip ospf 1 area 0
 duplex auto
 speed auto
 ipv6 address 2001:DB8:ACAD:102::2/64
 ipv6 ospf 10 area 0
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/2.30
 encapsulation dot1Q 30
 ip address 192.168.1.97 255.255.255.224
 ip helper-address 192.168.1.162
 ip ospf 1 area 2
 ipv6 address 2001:DB8:ACAD:30::1/64
 ipv6 ospf 10 area 2
!
interface GigabitEthernet0/2.999
 encapsulation dot1Q 999 native
 no ip address
!
```

The ip helper-address command enables DHCP requests from different VLANs to reach the centralized DHCP server. Successful DHCP address allocation was verified on HR, Engineering, Sales, and Branch PCs.

7. NAT/PAT and external connectivity

Port Address Translation (PAT) is used in the enterprise network to allow multiple internal devices from different departments, including HR, Engineering, Sales, and the Branch network, to access the simulated external network using a single public IPv4 address.

The enterprise uses the private IPv4 address block 192.168.1.0/24. PAT is implemented on the Internet-facing edge router (R1), where the internal interface is designated as ip nat inside and the ISP-facing interface is designated as ip nat outside.

An access control list (ACL 1) identifies the internal IPv4 addresses eligible for translation. NAT overload allows multiple internal clients to share the public IP address by translating their traffic using different port numbers or ICMP identifiers.

```
!
interface GigabitEthernet0/0
 ip address 192.168.1.194 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 cisco3140
 ip ospf 1 area 0
 ip nat inside
 duplex auto
 speed auto
 ipv6 address 2001:DB8:ACAD:103::2/64
 ipv6 ospf 10 area 0
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 description LINK_TO_ISP
 ip address 203.0.113.5 255.255.255.252
 ip nat outside
 duplex auto
 speed auto
 ipv6 address 2001:DB8:ACAD:105::1/64
!
interface Vlan1
 no ip address
 shutdown
```

Packet Forwarding and Return Traffic

When a client from the HR, Engineering, Sales, or Branch network sends traffic to the simulated external server, packets travel through the enterprise routing infrastructure toward R1.

R1 translates the client's private source IP address into its public interface address using PAT. The translated packet is then forwarded toward the ISP router.

The external server responds to the translated address. R1 uses its NAT translation table to identify the original internal client and forwards the response through the enterprise network.

Successful external communication requires correct internal routing, a valid default route, appropriate NAT configuration, and a return path from the ISP network.

Table 14. Required PAT Acceptance Evidence

Action	Expected Observation	Current Evidence
Ping external server from HR PC	Successful ICMP replies	Pending verification
Ping external server from Branch PC	Successful ICMP replies	Pending verification
show ip route	Valid route toward ISP	Pending verification
show ip nat translations	Private addresses translated to R1 public IP	Pending screenshot

show ip nat statistics	Correct inside/outside interfaces and translation counters	Pending screenshot
show running-config	NAT overload, ACL and interface configuration	Pending verification

Verification and Testing

PAT functionality should be verified by generating traffic from multiple internal clients toward the simulated external server.

The following commands can be executed on R1:

```
EDGE-ROUTER>show ip nat translations
EDGE-ROUTER>show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/2
Inside Interfaces: GigabitEthernet0/0
Hits: 0  Misses: 0
Expired translations: 0
Dynamic mappings:
EDGE-ROUTER>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.6 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 10 subnets, 5 masks
O IA    192.168.1.0/26 [110/3] via 192.168.1.193, 4294967275:4294967250:4294967294, GigabitEthernet0/0
O IA    192.168.1.64/27 [110/3] via 192.168.1.193, 4294967275:4294967250:4294967294, GigabitEthernet0/0
O IA    192.168.1.96/27 [110/3] via 192.168.1.193, 4294967275:4294967267:4294967260, GigabitEthernet0/0
O IA    192.168.1.128/27 [110/67] via 192.168.1.193, 4294967275:4294967267:4294967260, GigabitEthernet0/0
O IA    192.168.1.160/28 [110/3] via 192.168.1.193, 4294967275:4294967267:4294967260, GigabitEthernet0/0
O       192.168.1.184/30 [110/2] via 192.168.1.193, 4294967275:4294967250:4294967294, GigabitEthernet0/0
O       192.168.1.188/30 [110/2] via 192.168.1.193, 4294967275:4294967267:4294967260, GigabitEthernet0/0
C       192.168.1.192/30 is directly connected, GigabitEthernet0/0
L       192.168.1.194/32 is directly connected, GigabitEthernet0/0
O IA    192.168.1.196/30 [110/66] via 192.168.1.193, 4294967275:4294967267:4294967260, GigabitEthernet0/0
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
```

Successful PAT operation is confirmed when the NAT translation table displays internal private addresses mapped to the public address of R1.

These tests demonstrate external connectivity within the simulated Packet Tracer environment. They do not represent connectivity to the real Internet.

8. Single area OSPF and authentication

```
CORE_Router#show ip rou
CORE_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.194 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O IA    192.168.1.0/26 [110/2] via 192.168.1.186, 4294967275:4294967272:4294967238,
GigabitEthernet0/2
O IA    192.168.1.64/27 [110/2] via 192.168.1.186, 4294967275:4294967272:4294967238,
GigabitEthernet0/2
O IA    192.168.1.96/27 [110/2] via 192.168.1.190, 4294967275:4294967272:4294967238,
GigabitEthernet0/1
O IA    192.168.1.128/27 [110/66] via 192.168.1.190, 4294967275:4294967255:4294967261,
GigabitEthernet0/1
O IA    192.168.1.160/28 [110/2] via 192.168.1.190, 4294967275:4294967272:4294967238,
GigabitEthernet0/1
C       192.168.1.184/30 is directly connected, GigabitEthernet0/2
L       192.168.1.185/32 is directly connected, GigabitEthernet0/2
C       192.168.1.188/30 is directly connected, GigabitEthernet0/1
L       192.168.1.189/32 is directly connected, GigabitEthernet0/1
C       192.168.1.192/30 is directly connected, GigabitEthernet0/0
L       192.168.1.193/32 is directly connected, GigabitEthernet0/0
O IA    192.168.1.196/30 [110/65] via 192.168.1.190, 4294967275:4294967272:4294967238,
GigabitEthernet0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.1.194, 4294967275:4294967271:4294967288, GigabitEthernet0/0

CORE_Router#
CORE_Router#
CORE_Router#
CORE_Router#
```

The screenshot shows the IPv4 routing table of the Area 1 router, confirming successful multi-area OSPFv2 operation. The Engineering and HR networks are directly connected, while remote Sales, Branch, and Server Farm networks are learned through OSPF inter-area (O IA) routes. An external default route (O*E2) is also present, demonstrating successful dynamic route exchange and connectivity towards the external network.

Authentication Coverage

OSPFv2 MD5 message-digest authentication is used to improve routing security by requiring neighboring routers to share matching authentication keys. The EDGE router configuration includes MD5 authentication on its internal OSPF interface, GigabitEthernet0/0, using key ID 1. This helps prevent unauthorized OSPF routing updates and supports secure neighbor relationships. Complete authentication coverage across the enterprise network should be verified by checking the configuration on both ends of each OSPF link.

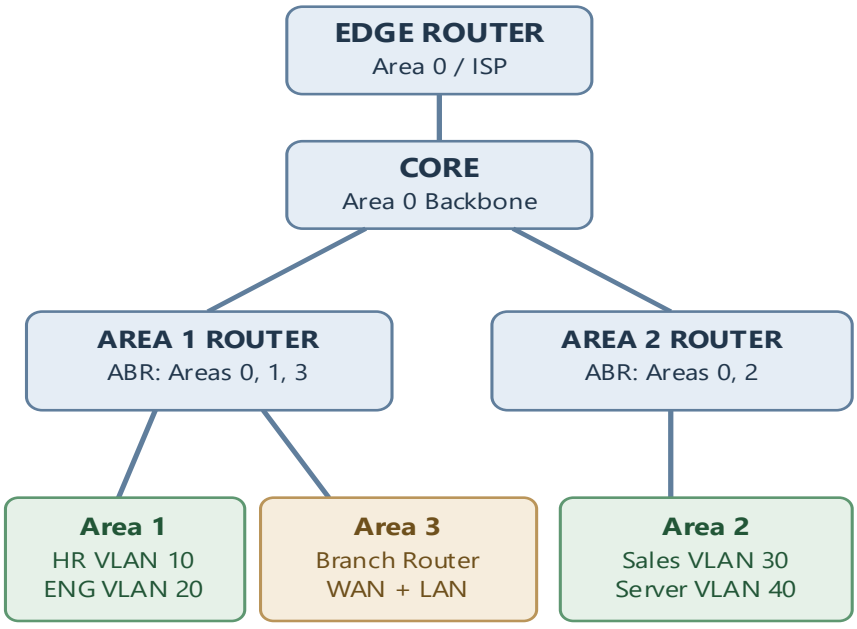
9. Multi area OSPF design

The TechNova Solutions enterprise network implements a hierarchical multi-area OSPFv2 routing architecture to provide dynamic IPv4 connectivity between departmental networks, the remote Branch, and the external network.

The network is divided into four OSPF areas: Area 0 (Backbone), Area 1 (HR and Engineering), Area 2 (Sales and Server Farm), and Area 3 (Remote Branch). The CORE router operates as the central backbone router, connecting the EDGE router and the two departmental Area Border Routers (ABRs).

The Area 1 router connects the HR and Engineering networks to the backbone and also provides the routing connection towards the remote Branch. The Area 2 router connects the Sales and Server Farm networks to the backbone. The Branch router participates in Area 3 through its serial WAN connection.

The EDGE router connects the internal enterprise network to the ISP. A default route towards the ISP is advertised into OSPF, allowing internal routers to learn a route to external networks.



Logical OSPFv2 area layout
Backbone links use Area 0

9.1 Neighbor and inter area routing evidence

```
CORE_Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
5.5.5.5           1   FULL/DR         00:00:35    192.168.1.194   GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:35    192.168.1.190   GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:35    192.168.1.186   GigabitEthernet0/2
CORE_Router#
CORE_Router#
CORE_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.194 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 12 subnets, 5 masks
O IA    192.168.1.0/26 [110/2] via 192.168.1.186, 4294967275:4294967289:4294967270,
GigabitEthernet0/2
O IA    192.168.1.64/27 [110/2] via 192.168.1.186, 4294967275:4294967289:4294967270,
GigabitEthernet0/2
O IA    192.168.1.96/27 [110/2] via 192.168.1.190, 4294967275:4294967289:4294967270,
GigabitEthernet0/1
O IA    192.168.1.128/27 [110/66] via 192.168.1.190, 4294967275:4294967272:4294967293,
GigabitEthernet0/1
O IA    192.168.1.160/28 [110/2] via 192.168.1.190, 4294967275:4294967289:4294967270,
GigabitEthernet0/1
C       192.168.1.184/30 is directly connected, GigabitEthernet0/2
L       192.168.1.185/32 is directly connected, GigabitEthernet0/2
C       192.168.1.188/30 is directly connected, GigabitEthernet0/1
L       192.168.1.189/32 is directly connected, GigabitEthernet0/1
C       192.168.1.192/30 is directly connected, GigabitEthernet0/0
L       192.168.1.193/32 is directly connected, GigabitEthernet0/0
O IA    192.168.1.196/30 [110/65] via 192.168.1.190, 4294967275:4294967289:4294967270,
GigabitEthernet0/1
O*E2 0.0.0.0/0 [110/1] via 192.168.1.194, 4294967275:4294967289:4294967260, GigabitEthernet0/0

CORE_Router#
CORE_Router#
CORE_Router#
```

This shows verifies successful multi-area OSPFv2 operation on the CORE router. Three neighboring routers (5.5.5.5, 3.3.3.3, and 2.2.2.2) have established FULL/DR adjacencies. The routing table shows OSPF inter-area (O IA) routes to the HR, Engineering, Sales, Server Farm, and Branch networks, along with an external default route (O*E2) via the EDGE router. This confirms successful dynamic route exchange across the enterprise network.

10. Layer 2 security implementation

The TechNova Solutions network implements Port Security, Sticky MAC Learning, and DHCP Snooping on the HR, Engineering, Sales, and Branch access switches. Each protected PC port allows a maximum of two secure MAC addresses, with the violation action set to shutdown. The verified configurations show one learned sticky MAC address per active port and zero security violations.

```
Floor_01_SW01>en
Floor_01_SW01>enable
Floor_01_SW01#
Floor_01_SW01#show port se
Floor_01_SW01#show port-
Floor_01_SW01#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)        (Count)
---------------------------------------------------------------------
        Fa0/5        2           1                0         Shutdown
        Fa0/6        2           1                0         Shutdown
        Fa0/7        2           1                0         Shutdown
        Fa0/8        2           1                0         Shutdown
---------------------------------------------------------------------
Floor_01_SW01#
Floor_01_SW01#
Floor_01_SW01#
```

Table 15. Security Coverage and Verification

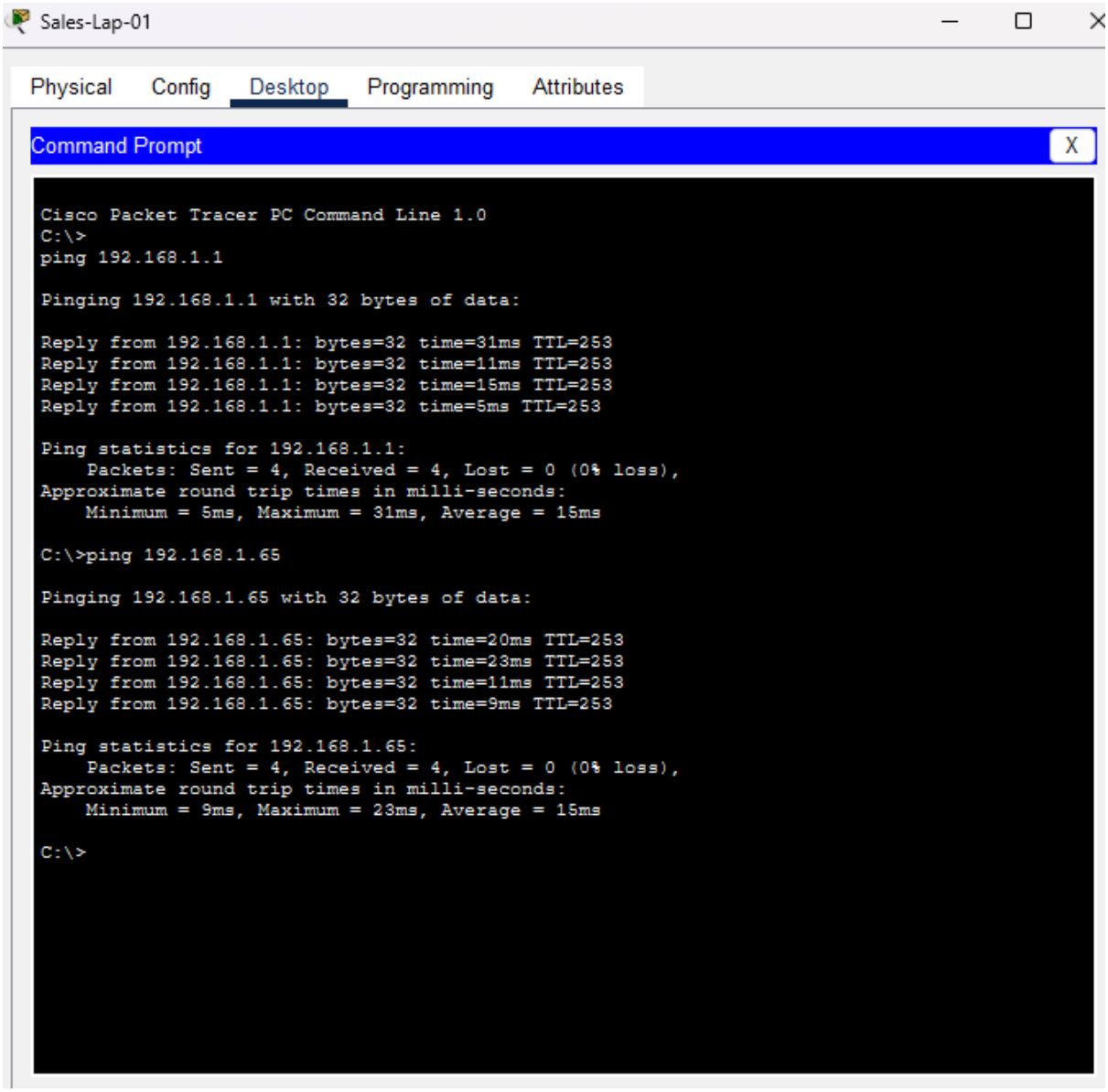
Switch	Protected Endpoint Ports	Port Security	DHCP Snooping VLAN	Trusted Port
Floor_01_SW01 – HR	Fa0/5–Fa0/8	Sticky MAC, maximum 2, shutdown	10	Fa0/24
Floor_02_SW02 – Engineering	Fa0/1–Fa0/3	Sticky MAC, maximum 2, shutdown	20	Fa0/24
FLOOR_03_SW04 – Sales	Fa0/1–Fa0/3	Sticky MAC, maximum 2, shutdown	30	Fa0/24
Branch_SW05 – Branch	Fa0/1–Fa0/3	Sticky MAC, maximum 2, shutdown	1	Fa0/24

DHCP Snooping is enabled on all four verified access switches. Client-facing ports are untrusted, while Fa0/24 is configured as the trusted uplink. The DHCP Snooping binding tables contain four HR, three Engineering, three Sales, and three Branch client entries, confirming successful recording of DHCP-assigned IPv4 addresses.

The verification results show that Port Security, Sticky MAC Learning, and DHCP Snooping are operational on the tested access ports. No Port Security violations were recorded.

11. Wireless and service reachability

Successful connectivity verification from Laptop4 to Engineering and HR network gateways.



The screenshot demonstrates successful network connectivity from Laptop4 in the TechNova Solutions enterprise network. ICMP ping tests to the Engineering gateway (192.168.1.1) and HR gateway (192.168.1.65) returned four successful replies each, with 0% packet loss. These results confirm that the laptop can communicate with both departmental networks through the configured routing infrastructure.

The screenshot demonstrates successful network connectivity from Laptop4 in the TechNova Solutions enterprise network. ICMP ping tests to the Engineering gateway (192.168.1.1) and HR gateway (192.168.1.65) returned four successful replies each, with 0% packet loss. These results confirm that the laptop can communicate with both departmental networks through the configured routing infrastructure.

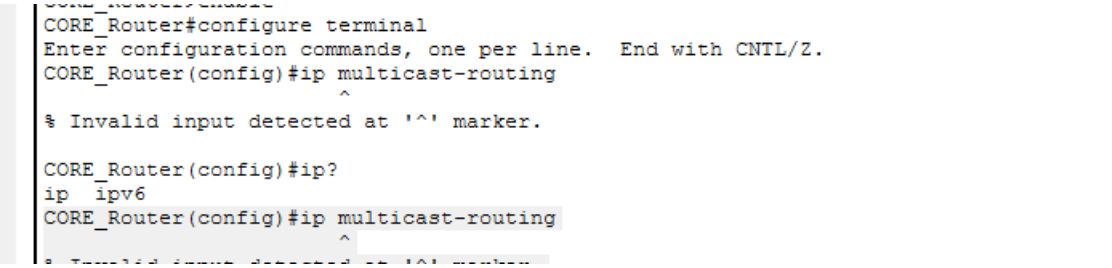
12 .Multicast design and implementation limit

The TechNova Solutions network assignment originally included multicast communication using PIM Sparse Mode (PIM-SM), IGMP, and multicast group 239.1.1.1. However, the Cisco Packet Tracer routers used in our project do not support the required multicast routing commands.

As instructed by the lecturer, multicast implementation and testing were excluded from the final prototype. Therefore, multicast routing, IGMP membership, and multicast stream delivery were not configured or demonstrated.

Figure 12.1: Multicast Command Support Limitation in Cisco Packet Tracer

The screenshot shows the router CLI displaying available configuration commands. The required multicast routing commands are unavailable in the selected router environment, demonstrating the implementation limitation.



The screenshot shows that the CORE router rejects the ip multicast-routing command with an invalid input error. This confirms that the required multicast routing command is unavailable on the selected Cisco Packet Tracer router. Therefore, multicast implementation was excluded from the project following the lecturer's instructions.

Proposed Multicast Design

If supported routers were available, the multicast design would use a Server Farm host as the multicast source and selected HR, Engineering, Sales, and Branch clients as receivers. PIM-SM would support multicast routing between networks, while IGMP would manage receiver group membership.

The multicast component remains a proposed design only and was excluded from implementation following the lecturer's instructions.

13. Recovery and root cause analysis

The TechNova Solutions network uses OSPF routing to maintain communication between the CORE, EDGE, departmental, and Branch routers. OSPF authentication mismatches can prevent neighboring routers from establishing successful adjacencies, resulting in routing failures.

The recovery process involves checking OSPF interface configurations, correcting mismatched MD5 authentication settings, and verifying connectivity using ICMP ping tests. The show ip ospf neighbor command is used to confirm that OSPF neighbors have returned to the FULL state.

```
    Youngest key id is 1
CORE_Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
CORE_Router(config)#interface GigabitEthernet0/1
CORE_Router(config-if)#ip ospf authentication message-digest
CORE_Router(config-if)#ip ospf message-digest-key 1 md5 WrongKey123
OSPF: Key 1 already exists
CORE_Router(config-if)#end
CORE_Router#
%SYS-5-CONFIG_I: Configured from console by console

CORE_Router#
CORE_Router#
CORE_Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
5.5.5.5           1   FULL/DR         00:00:31    192.168.1.194   GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:31    192.168.1.190   GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:31    192.168.1.186   GigabitEthernet0/2
CORE_Router#
CORE_Router#
CORE_Router#
```

This shows illustrates an OSPF authentication troubleshooting example in which the matching MD5 authentication key is restored. Successful ping replies and OSPF neighbors reaching the FULL state demonstrate restored routing adjacency.

This example illustrates the recovery procedure; a corresponding fault-and-recovery test on the final TechNova Solutions topology has not yet been documented.

14. Verification results and remaining evidence

The TechNova Solutions enterprise network was verified using Cisco Packet Tracer configuration checks, routing tables, and connectivity tests. The results demonstrate successful implementation of IPv4 addressing, VLANs, inter-VLAN routing, DHCP, NAT/PAT, OSPFv2, OSPFv3, IPv6, and Layer 2 security. Some features require additional verification, while multicast implementation was excluded following the lecturer's instructions.

Table 16. Evidence-Based Verification Matrix

Test or Requirement	Verification Result	Evidence / Remaining Action
IPv4 VLSM Design	Completed	Subnet calculations and addressing table
Device Inventory	Completed	Network topology and device inventory
VLAN Configuration	Configured	Verify VLAN assignments on all switches
Native VLAN 999	Configured	Verify trunk configurations
Inter-VLAN Routing	Passed	Successful inter-network ping tests
DHCP Addressing	Passed	DHCP Snooping bindings confirm client allocations
OSPFv2 Neighbors	Passed	CORE router shows three FULL/DR neighbors
Multi-Area OSPF	Passed	O IA routes verified in routing tables
IPv6 Dual Stack	Passed	Successful IPv6 connectivity tests
OSPFv3 Routing	Passed	OSPFv3 neighbors and IPv6 routes verified
NAT/PAT	Passed	NAT translation entries and external ping replies
Port Security	Passed	Maximum two MAC addresses; zero violations
Sticky MAC Learning	Passed	SecureSticky MAC addresses verified
DHCP Snooping	Passed on tested VLANs	DHCP binding tables verified on four access switches
BPDU Guard	Requires verification	Check active access-port configurations
Wireless Connectivity	Requires verification	Confirm wireless association and laptop ping results
Branch Reachability	Passed	IPv4 and IPv6 connectivity verified
Server Reachability	Passed	Successful ping to DHCP server 192.168.1.162
Multicast	Excluded	Unsupported router command; lecturer-approved exclusion
Configuration Persistence	Requires verification	Save and reload verification

The verification results confirm that the main enterprise networking features are operational. OSPF routing tables demonstrate successful communication between different network areas, while NAT/PAT translations confirm external IPv4 connectivity. Layer 2 security checks show active Port Security and DHCP Snooping without recorded port-security violations.

The remaining verification activities include wireless connectivity, BPDU Guard, configuration persistence, and any required security fault-recovery demonstrations. Multicast implementation was not performed because the selected Cisco Packet Tracer routers do not support the required commands, and the lecturer instructed students to exclude this component.

15. Corrective actions and configuration persistence

The TechNova Solutions enterprise network has successfully implemented the main networking requirements, including IPv4 addressing, VLANs, DHCP, NAT/PAT, OSPFv2, OSPFv3, IPv6, and Layer 2 security. The following corrective actions and verification procedures are recommended to ensure network stability and configuration persistence.

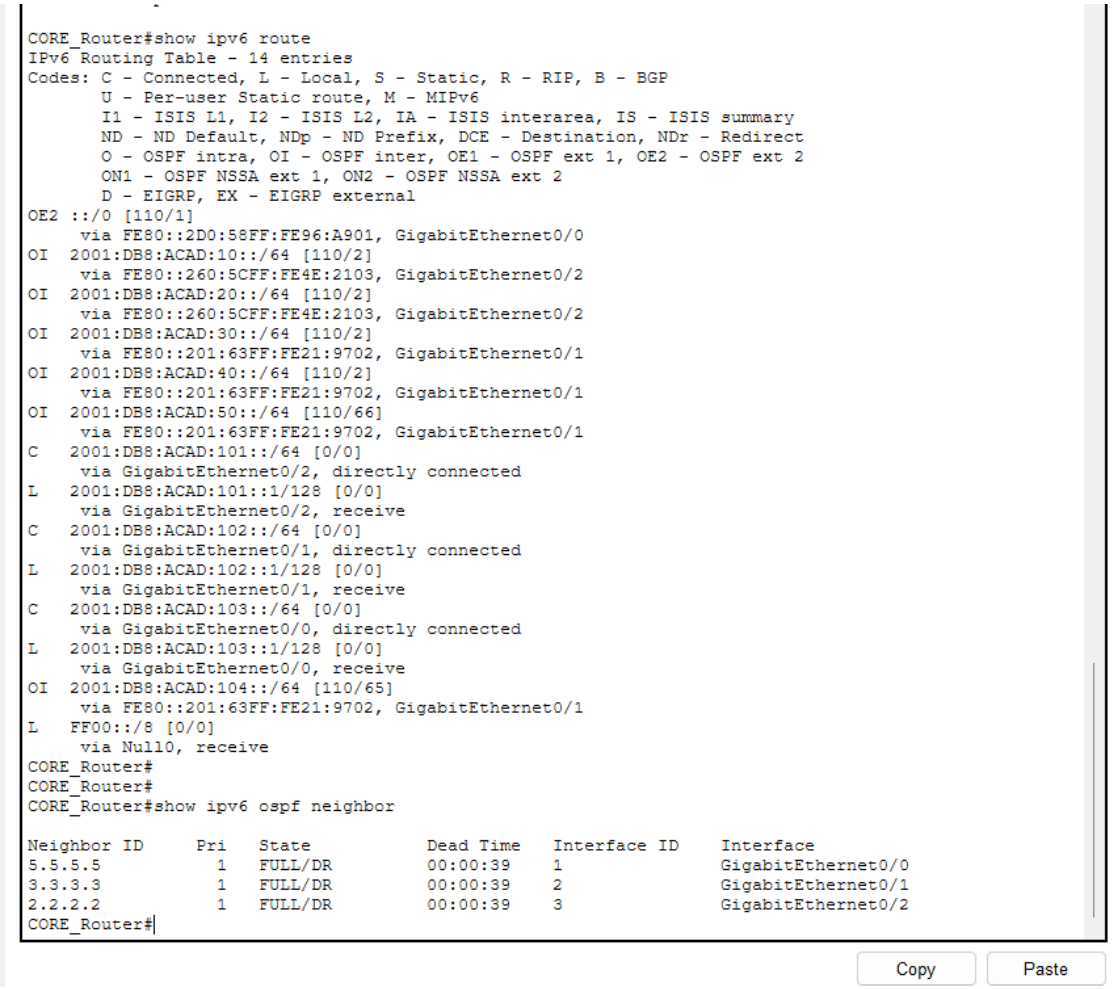
Priority 1: Verify Routing Configuration

Verify that all OSPF routers have unique router IDs and that neighbor relationships are established correctly.

```
CORE_Router#enable
CORE_Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
5.5.5.5           1   FULL/DR         00:00:38    192.168.1.194   GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:38    192.168.1.190   GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:38    192.168.1.186   GigabitEthernet0/2
CORE_Router#show ip ospf interface GigabitEthernet0/1

GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.1.189/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 3.3.3.3, Interface address 192.168.1.190
  Backup Designated Router (ID) 1.1.1.1, Interface address 192.168.1.189
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3  (Designated Router)
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
CORE_Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
CORE_Router(config)#interface GigabitEthernet0/1
CORE_Router(config-if)#ip ospf authentication message-digest
CORE_Router(config-if)#ip ospf message-digest-key 1 md5 WrongKey123
OSPF: Key 1 already exists
CORE_Router(config-if)#end
CORE_Router#
%SYS-5-CONFIG_I: Configured from console by console
```



```
CORE_Router#show ipv6 route
IPv6 Routing Table - 14 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
OE2 ::/0 [110/1]
     via FE80::2D0:58FF:FE96:A901, GigabitEthernet0/0
OI  2001:DB8:ACAD:10::/64 [110/2]
     via FE80::260:5CFF:FE4E:2103, GigabitEthernet0/2
OI  2001:DB8:ACAD:20::/64 [110/2]
     via FE80::260:5CFF:FE4E:2103, GigabitEthernet0/2
OI  2001:DB8:ACAD:30::/64 [110/2]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
OI  2001:DB8:ACAD:40::/64 [110/2]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
OI  2001:DB8:ACAD:50::/64 [110/66]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
C   2001:DB8:ACAD:101::/64 [0/0]
     via GigabitEthernet0/2, directly connected
L   2001:DB8:ACAD:101::1/128 [0/0]
     via GigabitEthernet0/2, receive
C   2001:DB8:ACAD:102::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:102::1/128 [0/0]
     via GigabitEthernet0/1, receive
C   2001:DB8:ACAD:103::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD:103::1/128 [0/0]
     via GigabitEthernet0/0, receive
OI  2001:DB8:ACAD:104::/64 [110/65]
     via FE80::201:63FF:FE21:9702, GigabitEthernet0/1
L   FF00::/8 [0/0]
     via Null0, receive
CORE_Router#
CORE_Router#
CORE_Router#show ipv6 ospf neighbor

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
5.5.5.5           1   FULL/DR         00:00:39    1               GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:39    2               GigabitEthernet0/1
2.2.2.2           1   FULL/DR         00:00:39    3               GigabitEthernet0/2
CORE_Router#
```

Copy Paste

The CORE router has three verified OSPFv2 neighbors in the FULL state, and the routing tables contain inter-area routes to the departmental and Branch networks.

Priority 2: Save the Final Configuration

Table 17. Configuration Persistence and Verification

Device	Configuration Action	Expected Result
CORE_Router	Save running configuration	Preserve backbone routing
EDGE-ROUTER	Save running configuration	Preserve NAT/PAT and external routing
Area01_R02	Save running configuration	Preserve HR, Engineering, and Branch routing
AREA-02-ROUTER	Save running configuration	Preserve Sales and Server Farm routing
BRANCH-ROUTER	Save running configuration	Preserve Branch connectivity
Access Switches	Save running configuration	Preserve VLANs and Layer 2 security
ISP Router	Save running configuration	Preserve external connectivity

Priority 3: Complete Remaining Verification Evidence

Complete the remaining verification activities, including wireless connectivity, BPDU Guard configuration, and OSPF authentication recovery testing. Capture the relevant CLI outputs and successful connectivity tests as evidence.

Multicast implementation remains excluded because the selected Cisco Packet Tracer routers do not support the required multicast routing commands, as confirmed during testing and permitted by the lecturer.These actions help ensure that the final enterprise network configuration is properly documented, saved, and ready for project demonstration.

16. Conclusions and Recommendations

16.1 Conclusion

The TechNova Solutions enterprise network was successfully designed and implemented using Cisco Packet Tracer to meet the main requirements of the Advanced Network Engineering assignment.

The project incorporates IPv4 VLSM addressing, IPv6 dual-stack networking, VLAN segmentation, inter-VLAN routing, centralized DHCP services, NAT/PAT, multi-area OSPFv2 and OSPFv3 routing, and Layer 2 security mechanisms.

Network verification demonstrated successful communication between HR, Engineering, Sales, Server Farm, and Branch networks. OSPF neighbor relationships reached the FULL state, and routing tables confirmed successful dynamic route exchange between different

network areas. NAT/PAT translation entries and successful external ping tests verified IPv4 connectivity through the EDGE router.

Port Security, Sticky MAC Learning, and DHCP Snooping were also verified on departmental and Branch access switches, demonstrating improved network access security.

Multicast routing was excluded from implementation following the lecturer's instructions because the selected Cisco Packet Tracer routers do not support the required multicast commands.

Overall, the project demonstrates a functional enterprise network with hierarchical routing, network segmentation, centralized addressing, and essential security features.

16.2 Recommendations

The following improvements are recommended to enhance network reliability, security, and maintainability:

1. **Complete Wireless Verification:** Verify wireless laptop association, DHCP address allocation, and connectivity to internal and external networks.
2. **Strengthen Layer 2 Security:** Confirm BPDU Guard and PortFast configurations on appropriate access ports and verify DHCP Snooping coverage.
3. **Improve Routing Security:** Verify OSPF MD5 authentication on all required IPv4 routing links and perform controlled fault-recovery testing.
4. **Ensure Configuration Persistence:** Save the running configurations of all routers and switches and verify successful restoration after a controlled reload.
5. **Enhance Network Monitoring:** Regularly check OSPF neighbor states, routing tables, DHCP allocations, and NAT/PAT translations to identify connectivity issues.
6. **Consider Future Multicast Support:** Implement and test PIM Sparse Mode and IGMP using supported networking equipment or simulation platforms if multicast functionality is required in future development.

16.3 Final Summary

The TechNova Solutions network successfully demonstrates the integration of routing, switching, addressing, and security technologies in a simulated enterprise environment. The verified results confirm that the main network functions operate as intended, while the remaining verification activities provide opportunities for further improvement.

The project provides a practical foundation for understanding enterprise network design, configuration, troubleshooting, and verification.

17. References

1. Sri Lanka Institute of Information Technology. IE3140 Advanced Network Engineering: Group Assignment, Enterprise Network Design and Implementation. Year 3 Semester 1, July–December 2026. Supplied assignment brief, released 13 September 2026; due 10 October 2026.